

## Question Bank

### Unit V: Automation in Marine Industry (12 Hrs.)

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**Syllabus:** Concepts in manufacturing and automation, reasons for automating; types of production and types of automation, automation strategies, levels of automation in shipbuilding; digital shipyards; robots in welding, blasting, heavy lifting and other tasks.

**Part A** 10 multiple-choice questions  $10 \times 1 = 10$  marks

**Part B** 5 short-answer questions  $5 \times 2 = 10$  marks

**Part C** 3 long-answer questions  $3 \times 10 = 30$  marks

#### Part A: Multiple-Choice Questions

1 mark each

**Q1.** Automation in which the sequence is fixed by the equipment design and used for very high volumes is called

- (a) flexible automation
- (b) programmable automation
- (c) fixed automation
- (d) manual operation

**Answer:** (c) Fixed (hard) automation.

**Q2.** Shipbuilding belongs to which type of production?

- (a) Low-quantity (job-shop) production
- (b) Mass production
- (c) Continuous flow production
- (d) Batch production of millions

**Answer:** (a) Low-quantity, one-of-a-kind (job-shop) production.

**Q3.** The USA principle of automation stands for

- (a) Use, Save, Apply
- (b) Unite, Standardise, Assemble
- (c) Update, Store, Analyse
- (d) Understand, Simplify, Automate

**Answer:** (d) Understand, Simplify, Automate.

**Q4.** In modern shipbuilding, the hull is built in large sections called

- (a) frames
- (b) blocks
- (c) modules of paint
- (d) stringers

**Answer:** (b) Blocks (hull block construction method).

**Q5.** The plant layout in which the product stays in one place and workers and equipment come to it is

- (a) fixed-position layout
- (b) process layout
- (c) product layout
- (d) cellular layout

**Answer:** (a) Fixed-position layout — e.g. hull erection in the building dock.

**Q6.** An automatic panel line in a shipyard is an example of a

- (a) fixed-position layout
- (b) process layout
- (c) product (flow-line) layout
- (d) random layout

**Answer:** (c) Product (flow-line) layout — panels move from station to station.

**Q7.** A digital twin is

- (a) a backup computer
- (b) a virtual replica of a product or process updated with real data
- (c) a pair of identical ships
- (d) a type of welding robot

**Answer:** (b) A virtual replica updated with real-time data.

**Q8.** A welding robot finds the real position of a seam by touching the plates with the welding wire. This is called

- (a) arc sensing
- (b) ultrasonic testing
- (c) laser scanning
- (d) touch sensing

**Answer:** (d) Touch sensing.

**Q9.** SPMT, used for moving heavy hull blocks, stands for

- (a) Special Purpose Mobile Tractor
- (b) Shipyard Plate Moving Truck
- (c) Self-Propelled Modular Transporter
- (d) Single Point Mooring Trailer

**Answer:** (c) Self-Propelled Modular Transporter.

**Q10.** The IMO PSPC standard relates to

- (a) protective coatings of ballast tanks
- (b) welding of hull plates
- (c) crane load testing
- (d) engine emissions

**Answer:** (a) Performance Standard for Protective Coatings (e.g. dedicated seawater ballast tanks).

## Part B: Short-Answer Questions

2 marks each

**Q1.** Define manufacturing from the technological and economic points of view.

[2]

### Answer

**Technological:** application of physical and chemical processes to alter the geometry, properties or appearance of a starting material to make parts or products. **Economic:** transformation of materials into items of **greater value** by processing and assembly operations — manufacturing *adds value*.

**Q2.** List any four reasons for automating in shipbuilding.

[2]

### Answer

(i) Shortage of skilled welders and fitters; (ii) hazardous work — welding fumes in confined double-bottom cells, blasting dust, paint solvents; (iii) consistent weld and coating quality demanded by class and IMO rules; (iv) higher productivity and shorter delivery times to meet global price competition.

**Q3.** A stiffener-cutting station has a cycle time of 6 minutes per part and an availability of 0.9. Find the ideal and the effective production rate.

[2]

### Answer

Ideal  $R_p = 60/T_c = 60/6 = 10$  parts/h. Effective rate  $= A \times R_p = 0.9 \times 10 = 9$  parts/h.

**Q4.** What is a digital shipyard?

[2]

**Answer**

A digital shipyard (Shipyard 4.0) is one in which design, planning, production and after-sales share **one 3-D product model** and **real-time data** from sensors, RFID and machines, connected by networks and software (CAD/CAM, MES, ERP, digital twin) — Industry 4.0 applied to shipbuilding.

- Q5.** A 300 t hull block is lifted by two hooks A and B, 20 m apart. Its centre of gravity is 8 m from A. Find the load on each hook. [2]

**Answer**

Taking moments:  $R_A = W \frac{12}{20} = 300 \times 0.6 = 180 \text{ t}$ ;  $R_B = W \frac{8}{20} = 300 \times 0.4 = 120 \text{ t}$ . The hook nearer the CG carries more load.

**Part C: Long-Answer Questions****10 marks each**

- Q1.** Explain the types of automation with examples from shipbuilding. Discuss the ten strategies for automation and the automation migration strategy with a neat diagram. [10]

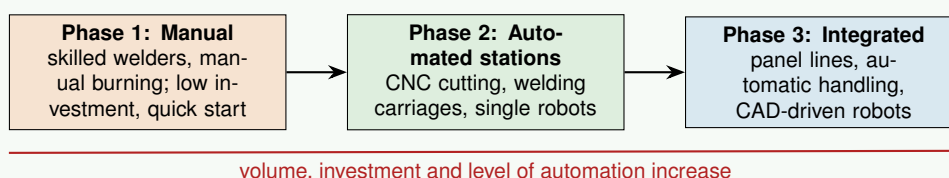
**Answer****(a) Types of automation**

| Type         | Features                                                                 | Shipyard example                                     |
|--------------|--------------------------------------------------------------------------|------------------------------------------------------|
| Fixed        | sequence fixed by equipment; high volume; high investment, low unit cost | plate shot-blasting and priming line                 |
| Programmable | new product = new program; batch production; set-up time lost            | CNC plasma cutting, robot welding of sub-assemblies  |
| Flexible     | mix of products with no change-over loss; programs made off-line         | panel line with robots programmed from the 3-D model |

Since every ship is different, shipyards rely mainly on **programmable and flexible** automation.

**(b) Ten strategies for automation**

1. Specialisation of operations — dedicated stiffener-welding gantry.
2. Combined operations — cut, mark and bevel in one CNC set-up.
3. Simultaneous operations — several welding heads on one panel.
4. Integration of operations — blasting, priming and cutting linked by conveyors.
5. Increased flexibility — robots programmed per panel from CAD.
6. Improved material handling — transporters, automated plate stock-yard.
7. On-line inspection — laser scanning of block dimensions.
8. Process control & optimisation — adaptive welding with seam tracking.
9. Plant operations control — MES tracking of blocks and pipe spools.
10. Computer-integrated manufacturing — one product model from design to delivery.

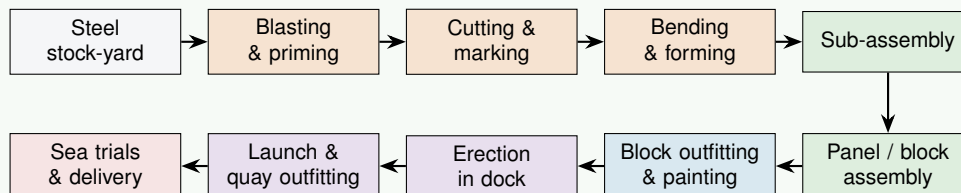
**(c) Automation migration strategy**

Automation is introduced in steps as demand and experience grow: each phase is proven before the next investment, reducing risk, and the same product-model data are reused at every phase.

**Q2.** Draw a flowchart of the shipbuilding production process. Explain the hull block construction method, zone outfitting, and the levels of automation at different production stages. [10]

### Answer

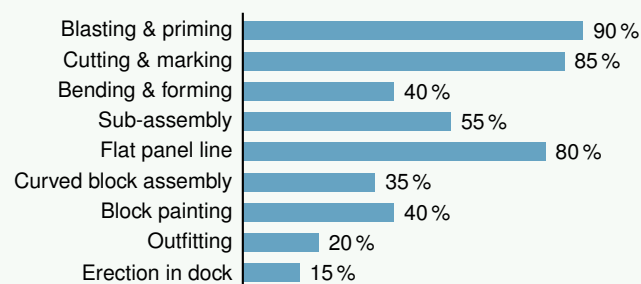
#### (a) Production process



**(b) Hull block construction method (HBCM):** the hull is divided into **blocks** (50–900 t) built in parallel in covered workshops from parts, sub-assemblies and panels, then lifted by gantry cranes and **erected** in the dock. Benefits: parallel work, better access and working positions (down-hand welding), shorter dock time.

**(c) Zone outfitting and zone painting:** pipes, cable trays, equipment foundations and machinery are fitted, and blocks are painted, **before** erection (on-block / on-unit outfitting). Work is done in the open, easy-to-reach block rather than in the finished ship, saving time and cost.

#### (d) Levels of automation by stage (indicative)

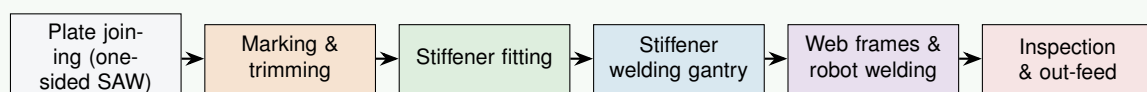


Automation is **highest** where work is flat, straight and repetitive (plate lines, cutting, flat panels) and **lowest** where work is three-dimensional, confined or unique (curved blocks, outfitting, erection). Levels range from manual → mechanised (welding tractor) → semi-automatic (portable robot) → automatic (gantry robots) → integrated (CAD-driven, MES-controlled).

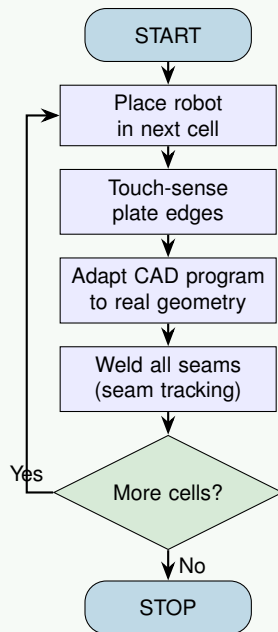
**Q3.** Explain the use of robots in shipyard welding with a neat diagram of an automatic panel line and a flowchart of the portable welding-robot cycle. Show, with an example, how robots improve productivity. Briefly mention robots for blasting, painting and heavy lifting. [10]

### Answer

#### (a) Automatic panel line



Flat panels move on roller conveyors from station to station (product layout). Multi-head gantry robots weld stiffeners on both sides at once, reducing distortion; weld paths come from the 3-D model (**off-line programming**).

**(b) Portable robot cycle**

**Explanation:** small portable robots are lifted by crane into the **double-bottom cells**. Touch sensing finds the real plate positions, the CAD-generated program is corrected, and all fillet welds are made with arc seam tracking. Weld data are logged for class-society records.

**(c) Productivity example:** 200 m of fillet weld,  $t = L/(v f)$ .

- **Manual** ( $v = 0.3$  m/min,  $f = 30\%$ ):  
 $t = 200/(0.3 \times 0.3) = 2222 \text{ min} \approx \mathbf{37.0 \text{ h}}$
- **Robot** ( $v = 0.5$  m/min,  $f = 80\%$ ):  
 $t = 200/(0.5 \times 0.8) = 500 \text{ min} \approx \mathbf{8.3 \text{ h}}$
- Time saved  $\approx \mathbf{77.5\%}$  — mainly from the higher arc-on time.

**(d) Other robots**

- **Blasting & painting:** magnetic crawlers with grit or UHP water-jet heads (dust vacuumed back); boom/gantry painting robots keep a constant stand-off, angle and speed, giving uniform DFT and higher transfer efficiency (e.g. 85% vs 60%), saving about 29% paint.
- **Heavy lifting:** Goliath gantry cranes with synchronised hoists and anti-sway control for block erection and turn-over; SPMTs with computer-controlled steering and hydraulic load levelling move blocks of several hundred tonnes.